

Chapter 9

Integral Calculus

9.1 Integrating $ax^n = f(x)$

Integration is the reverse process of differentiation. In Chapter 5, we saw that if $y = f(x) = ax^n$, then

$$\frac{dy}{dx} = f'(x) = anx^{n-1},$$

and that for any constant c , if $y = f(x) = ax^n + c$, then

$$\frac{dy}{dx} = f'(x) = anx^{n-1}.$$

Since integration is the reverse process of differentiation, we seek to find $y = f(x)$, given $\frac{dy}{dx} = f'(x)$. That is; if $\frac{dy}{dx} = f'(x) = kx^n$, then

$$y = f(x) = \frac{kx^{n+1}}{n+1} + c, \quad n \neq -1$$

and c is called a constant of integration.

Example 9.1.1. Find y for each of the following:

1. $\frac{dy}{dx} = 3x^4$

2. $\frac{dy}{dx} = 2x^{-5}$

3. $\frac{dy}{dx} = 6x + 2x^{-3} - 3x^{\frac{1}{2}}$

Solutions

1.

$$\begin{aligned}\frac{dy}{dx} &= 3x^4 \\ y &= \frac{3x^{4+1}}{4+1} + c \\ &= \frac{3}{5}x^5 + c.\end{aligned}$$

2.

$$\begin{aligned}\frac{dy}{dx} &= 2x^{-5} \\ y &= \frac{2x^{-5+1}}{-5+1} + c \\ &= -\frac{1}{2}x^{-4} + c.\end{aligned}$$

3.

$$\begin{aligned}\frac{dy}{dx} &= 6x + 2x^{-3} - 3x^{\frac{1}{2}} \\ y &= 3x^2 - x^{-2} - 2x^{\frac{3}{2}} + c\end{aligned}$$

9.2 Using the integral symbol

The integral of x^n is denoted by $\int x^n dx$ and the formula we have been using is:

$$\int kx^n dx = \frac{kx^{n+1}}{n+1} + c, \quad n \neq -1.$$

Example 9.2.1. Find:

1. $\int (x^{\frac{1}{2}} + 2x^3) dx$

2. $\int (x^{-\frac{3}{2}} + 2) dx$

Solution

1.

$$\int (x^{\frac{1}{2}} + 2x^3) dx = \frac{2}{3}x^{\frac{3}{2}} + \frac{1}{2}x^4 + c.$$

2.

$$\int (x^{-\frac{3}{2}} + 2) dx = -2x^{\frac{-1}{2}} + 2x + c.$$

9.2.2 Simplifying expression before integrating:

Here we will first simplify the expression into terms of the form x^n , $n \in \mathbb{R}$ before integrating.

Example 9.2.3. Find:

1. $\int (\frac{2}{x^3} - 3\sqrt{x}) dx$

2. $\int x(x^2 + \frac{2}{x}) dx$

3. $\int ((2x^2) + \frac{5+\sqrt{x}}{x^2}) dx$

Solution

1.

$$\begin{aligned} \int (\frac{2}{x^3} - 3\sqrt{x}) dx &= \int (2x^{-3} - 3x^{\frac{1}{2}}) dx \\ &= -x^{-2} - 2x^{\frac{3}{2}} + c \end{aligned}$$

2.

$$\begin{aligned} \int x(x^2 + \frac{2}{x}) dx &= \int (x^3 + 2) dx \\ &= \frac{1}{4}x^4 + 2x + c. \end{aligned}$$

3. HW

The following are some standard integrals

1.

$$\int \sin x dx = -\cos x + c$$

2.

$$\int \cos x dx = \sin x + c$$

3.

$$\int \frac{1}{x} dx = \ln x + c$$

4.

$$\int e^x dx = e^x + c$$

5.

$$\int a^x dx = \frac{a^x}{\ln a} + c$$

9.3 Finding the constant of integration

We know that if $\frac{dy}{dx} = f'(x) = kx^n$, then

$$y = f(x) = \frac{kx^{n+1}}{n+1} + c, \quad n \neq -1.$$

Given any point (x, y) where the curve $y = \frac{kx^{n+1}}{n+1} + c$ passes through, we can find the actual value of c .

Example 9.3.1. The curve with equation $y = f(x)$ passes through the point $(4, 5)$. Given that

$$f'(x) = \frac{x^2 - 2}{\sqrt{x}},$$

find the equation of the curve.

9.3.2 Definite Integrals

We need to be able to integrate functions within defined limits. This is called definite integration.

We recall that

$$\int 3x^4 dx = \frac{3}{5}x^5 + c,$$

where c is an arbitrary constant.

Let $f(x)$ be continuous function between $x = a$ and $x = b$ and let $F(x)$ be the integral of $f(x)$. Then the **definite integral** between the limits $x = a$ and $x = b$ is

$$\int_a^b (f(x)) \, dx = [F(x)]_a^b = F(b) - F(a),$$

where b is the upper limit and a is the lower limit of integration.

Example 9.3.3. Evaluate the following:

1.
$$\int_1^4 (2x - 3x^{\frac{1}{2}} + 1) \, dx$$

2.
$$\int_0^1 (x^{\frac{1}{2}} - 1)^2 \, dx$$

9.4 Area under a curve

You need to be able to find the area under a curve.

9.4.1 Area bounded by the x -axis and the curve above the x -axis

If you wish to find the area between a curve $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$, then

$$A = \int_a^b y \, dx.$$

Example 9.4.2. Find the area of the region bounded by the curve with equation $y = (4 - x)(x + 2)$ and the positive x - and y -axes.

9.4.3 Area bounded by the x -axis and the curve under the x -axis

If you wish to find the area between a curve $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$, then

$$A = - \int_a^b y \, dx.$$

Example 9.4.4. Find the area of the region bounded by the curve with equation $y = x(x - 3)$ and the x -axis.

The following example shows that great care must be taken if you are trying to find an area which straddles the x -axis such as the shaded region below.

Example 9.4.5. Sketch the curve with equation $y = x(x - 1)(x + 3)$ and find the area of the region bounded by the curve and the x -axis.

9.4.6 Area between a curve and a straight line

The figure below shows the graph of $y_1 = f(x)$ and $y_2 = g(x)$. The area between y_1 and y_2 is given by

$$A = \int_a^b (y_1 - y_2) dx.$$

Example 9.4.7. Sketch the graph of the curve $y = x(4 - x)$ and the line $y = x$. Hence find the area of the region bounded by the two functions.

9.5 Methods of integration

We now consider some methods of integrating functions which are not given in standard form, such as the ones we have looked at so far. The idea is to standardize the integrand so that we are able to integrate using known/standard results. Some of the known/standard integrals can easily be found on a standard integral table; on the internet or in most of the advanced Mathematics books.

9.5.1 Substitution method

Suppose we wish to find

$$\int 2x(x^2 + 3)^8 dx$$

or

$$\int \frac{\cos x}{\sin x} dx.$$

First, we observe that the two functions being multiplied or divided are related in some way. That is ; $\frac{d}{dx}(x^2 + 3) = 2x$ and $\frac{d}{dx}(\sin x) = \cos x$,

and so we use *substitution method*.

With this method, the substitution usually made is to let u be equal to $f(x)$ such that $f(u) du$ is a standard integral. It is known that integrals of the forms,

$$k \int [f(x)]^n f'(x) dx$$

and

$$k \int \frac{f'(x)}{[f(x)]^n} dx$$

where k and n are constants, can both be integrated by using substituting u for $f(x)$. The following example will help us appreciate this method.

Example 9.5.2. Integrate each of the following:

1.

$$\int 2x(x^2 + 3)^2 dx$$

2.

$$\int \frac{2x + 1}{3x^2 + 3x + 1} dx.$$

3.

$$\int 3x^2 e^{x^3} dx.$$

4.

$$\int \frac{\cos x}{\sin x} dx.$$

5.

$$\int \frac{\ln x}{x} dx.$$

9.5.3 Integration by parts

From the product rule of differentiation

$$\frac{d}{dx}(uv) = v \frac{du}{dx} + u \frac{dv}{dx},$$

where both u and v are functions of x , we get

$$u \frac{dv}{dx} = \frac{d}{dx}(uv) - v \frac{du}{dx}.$$

Integrating both sides with respect to x gives:

$$\int u dv = uv - \int v du.$$

This is called *integration by parts* formula and provides a method of integrating such products of simple functions as $\int x e^x dx$, $\int t^3 \sin t dt$, $\int e^x \cos x dx$ and $\int x^2 \ln x dx$.

Given a product of two terms to integrate, the initial choice is: which part to make equal to u and which part to make equal to dv . The choice must be such that the 'u part' becomes a constant after successive differentiation, and the 'dv part' can be integrated from standard integrals.

The following rule holds: If a product to be integrated contains an algebraic term (such as x , t^2 or 3θ), then this term is chosen as the 'u part', except when $\ln x$ term is involved; in this case $\ln x$ is chosen as the 'u part'.

We consider the examples below.

Example 9.5.4. Determine

1. $\int x \cos x dx$.
2. $\int 3te^{2t} dt$.
3. $\int x^2 \sin x dx$.

Example 9.5.5. Determine

1. $\int x \ln x dx$.
2. $\int e^{2x} \cos 4x dx$.

Solution

1. $\int x \ln x \, dx$. Here, our choice of u is the log. function. Thus, let $u = \ln x$ and $dv = x \, dx$. we have $\frac{du}{dx} = \frac{1}{x}$ and $v = \frac{x^2}{2}$. By

$$\int u \, dv = uv - \int v \, du,$$

we get

$$\begin{aligned}\int x \ln x \, dx &= \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \times \frac{1}{x} \, dx \\ &= \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx \\ &= \frac{x^2}{2} \ln x - \frac{x^2}{4} + c\end{aligned}$$

2. $\int e^{2x} \cos 4x \, dx$. Here, the choice of u does not matter as shown below; Let $u = e^{2x}$ and $dv = \cos 4x \, dx$. It follows that $du = 2e^{2x} \, dx$ and $v = \frac{1}{4} \sin 4x$. Again by integration by parts formula, we get

$$\begin{aligned}\int e^{2x} \cos 4x \, dx &= \frac{1}{4} e^{2x} \sin 4x - \int \frac{1}{4} \sin 4x \times 2e^{2x} \, dx \\ &= \frac{1}{4} e^{2x} \sin 4x - \frac{1}{2} \int e^{2x} \sin 4x \, dx.\end{aligned}$$

To integrate $\int e^{2x} \sin 4x$, We apply the method of integration by part again. Let $u = e^{2x}$ and $dv = \sin 4x \, dx$. It follows that $du = 2e^{2x} \, dx$ and $v = -\frac{1}{4} \cos 4x$. Thus

$$\int e^{2x} \sin 4x = -\frac{1}{4} e^{2x} \cos 4x + \frac{1}{2} \int e^{2x} \cos 4x \, dx.$$

Consequently,

$$\begin{aligned}
\int e^{2x} \cos 4x \, dx &= \frac{1}{4} e^{2x} \sin 4x - \int \frac{1}{4} \sin 4x \times 2e^{2x} \, dx \\
&= \frac{1}{4} e^{2x} \sin 4x - \frac{1}{2} \int e^{2x} \sin 4x \, dx \\
&= \frac{1}{4} e^{2x} \sin 4x - \frac{1}{2} \left(-\frac{1}{4} e^{2x} \cos 4x + \frac{1}{2} \int e^{2x} \cos 4x \, dx \right) \\
\int e^{2x} \cos 4x \, dx &= \frac{1}{4} e^{2x} \sin 4x + \frac{1}{8} e^{2x} \cos 4x - \frac{1}{4} \int e^{2x} \cos 4x \, dx.
\end{aligned}$$

Hence,

$$\begin{aligned}
\int e^{2x} \cos 4x \, dx + \frac{1}{4} \int e^{2x} \cos 4x \, dx &= \frac{1}{4} e^{2x} \sin 4x + \frac{1}{8} e^{2x} \cos 4x \\
\frac{5}{4} \int e^{2x} \cos 4x \, dx &= \frac{1}{4} e^{2x} \sin 4x + \frac{1}{8} e^{2x} \cos 4x \\
\int e^{2x} \cos 4x \, dx &= \frac{1}{5} e^{2x} \sin 4x + \frac{1}{10} e^{2x} \cos 4x + c.
\end{aligned}$$

9.5.6 Integration using partial fractions

The process of expressing a fraction in terms of simpler fractions called *partial fractions* was introduced in chapter 5. The following list summarizes the forms of partial fractions used

Denominator containing	Expression	Form of partial fraction
Linear factors	$\frac{f(x)}{(x+a)(x-b)(x+c)}$	$\frac{A}{x+a} + \frac{B}{x-b} + \frac{C}{x+c}$
Repeated linear factors	$\frac{f(x)}{(x+a)^3}$	$\frac{A}{x+a} + \frac{B}{(x+a)^2} + \frac{C}{(x+a)^3}$
Quadratic factors	$\frac{f(x)}{(ax^2+bx+c)(x+d)}$	$\frac{Ax+B}{(ax^2+bx+c)} + \frac{C}{x+d}$

The general idea here is to split the given rational function into its partial fractions, which can now be integrated using the methods dealt with earlier. In particular, we will employ the method of substitution. To that effect, we have the following.

Example 9.5.7. Find

1. $\int \frac{11-3x}{x^2+2x-3} dx.$

2. $\int \frac{2x^2-9x-35}{(x+1)(x-2)(x+3)} dx.$

3. $\int \frac{x^3-2x^2-4x-4}{x^2+x-2} dx.$

4. $\int \frac{2x+3}{(x-2)^3} dx.$

5. $\int \frac{3+6x+4x^2-2x^3}{x^2(x^2+3)} dx.$

Solution

1. $\int \frac{11-3x}{x^2+2x-3} dx.$ By Example 6.2.1, $\frac{11-3x}{x^2+2x-3} = \frac{2}{x-1} - \frac{5}{x+3}.$ By using the substitution method, it follows that

$$\begin{aligned}\int \frac{11-3x}{x^2+2x-3} dx &= \int \left(\frac{2}{x-1} - \frac{5}{x+3} \right) dx \\ &= \int \frac{2}{x-1} dx - \int \frac{5}{x+3} dx \\ &= 2 \ln |x-1| - 5 \ln |x+3| + c.\end{aligned}$$

2. HW

3. $\int \frac{x^3-2x^2-4x-4}{x^2+x-2} dx.$ By partial fraction decomposition,

$$\begin{aligned}\frac{x^3-2x^2-4x-4}{x^2+x-2} &= x-3 + \frac{x-10}{x^2+x-2} \\ &= x-3 + \frac{x-10}{(x+2)(x-1)} \\ &= x-3 + \frac{A}{x+2} + \frac{B}{x-1} \\ &= x-3 + \frac{4}{x+2} + \frac{-3}{x-1}.\end{aligned}$$

Consequently,

$$\begin{aligned}\int \frac{x^3 - 2x^2 - 4x - 4}{x^2 + x - 2} dx &= \int \left(x - 3 + \frac{4}{x+2} + \frac{-3}{x-1} \right) dx \\ &= \int x dx - \int 3 dx + \int \frac{4}{x+2} dx - \int \frac{3}{x-1} dx \\ &= \frac{x^2}{2} - 3x + 4 \ln |x+2| - 3 \ln |x-1| + c.\end{aligned}$$

4. HW

5. HW

Certain functions can be integrated by changing the variable of integration as shown below

Example 9.5.8. Integrate

1. $\int \frac{x}{\sqrt{x+1}} dx.$

2. $\int \frac{e^{2x}}{e^x+1} dx.$

Solution

1. $\int \frac{x}{\sqrt{x+1}} dx.$ Here, the method of substitution does not yield the intended result. We let $u = x + 1$ so that $du = dx$. It follows that $\int \frac{x}{\sqrt{x+1}} dx = \int \frac{x}{\sqrt{u}} du$. Since we are integrating with respect to u , we change the variable to u . But from $u = x + 1$, $x = u - 1$, and so

$$\begin{aligned}\int \frac{x}{\sqrt{x+1}} dx &= \int \frac{x}{\sqrt{u}} du \\ &= \int \frac{u-1}{\sqrt{u}} du \\ &= \int (u^{\frac{1}{2}} - u^{-\frac{1}{2}}) dx \\ &= \frac{2}{3} u^{\frac{3}{2}} - 2\sqrt{u} + c \\ &= \frac{2}{3} (x+1)^{\frac{3}{2}} - 2\sqrt{x+1} + c \\ &= \frac{2}{3} \sqrt{x+1} (x-2) + c.\end{aligned}$$

2. $\int \frac{e^{2x}}{e^x + 1} dx$. Let $u = e^x + 1$, so that $du = e^x dx$. Thus

$$\begin{aligned}\int \frac{e^{2x}}{e^x + 1} dx &= \int \frac{e^x \cdot e^x}{e^x + 1} dx \\&= \int \frac{u - 1}{u} du \\&= \int (1 - u^{-1}) du \\&= u - \ln |u| + c \\&= e^x + 1 - \ln |e^x + 1| + c \\&= e^x - \ln |e^x + 1| + k.\end{aligned}$$

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